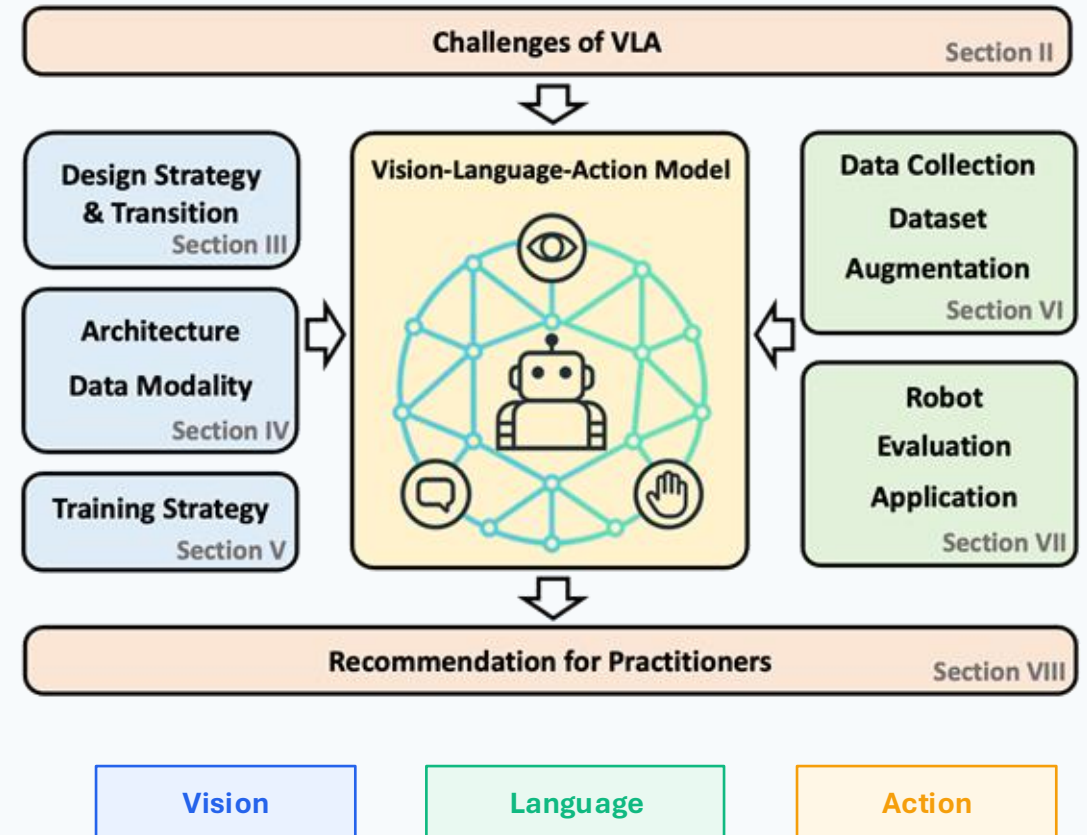


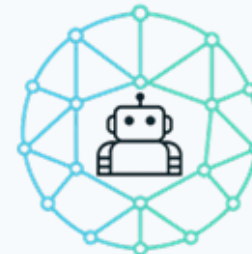
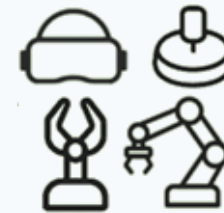
Vision-Language-Action models (VLAs): What, Why, How?

- VLAs is a system that takes **visual observations** and **natural language instructions** as required inputs and may incorporate additional sensory modalities. It produces **robot actions** by directly generating control commands.
- VLAs aim to enable **robots** to perform **a wider range of tasks** and facilitate effective transfer across **varying robotic embodiments**.
- Jointly learning **visual**, **linguistic**, and **action** modalities in an **end-to-end** framework with different architectural and training methodologies.



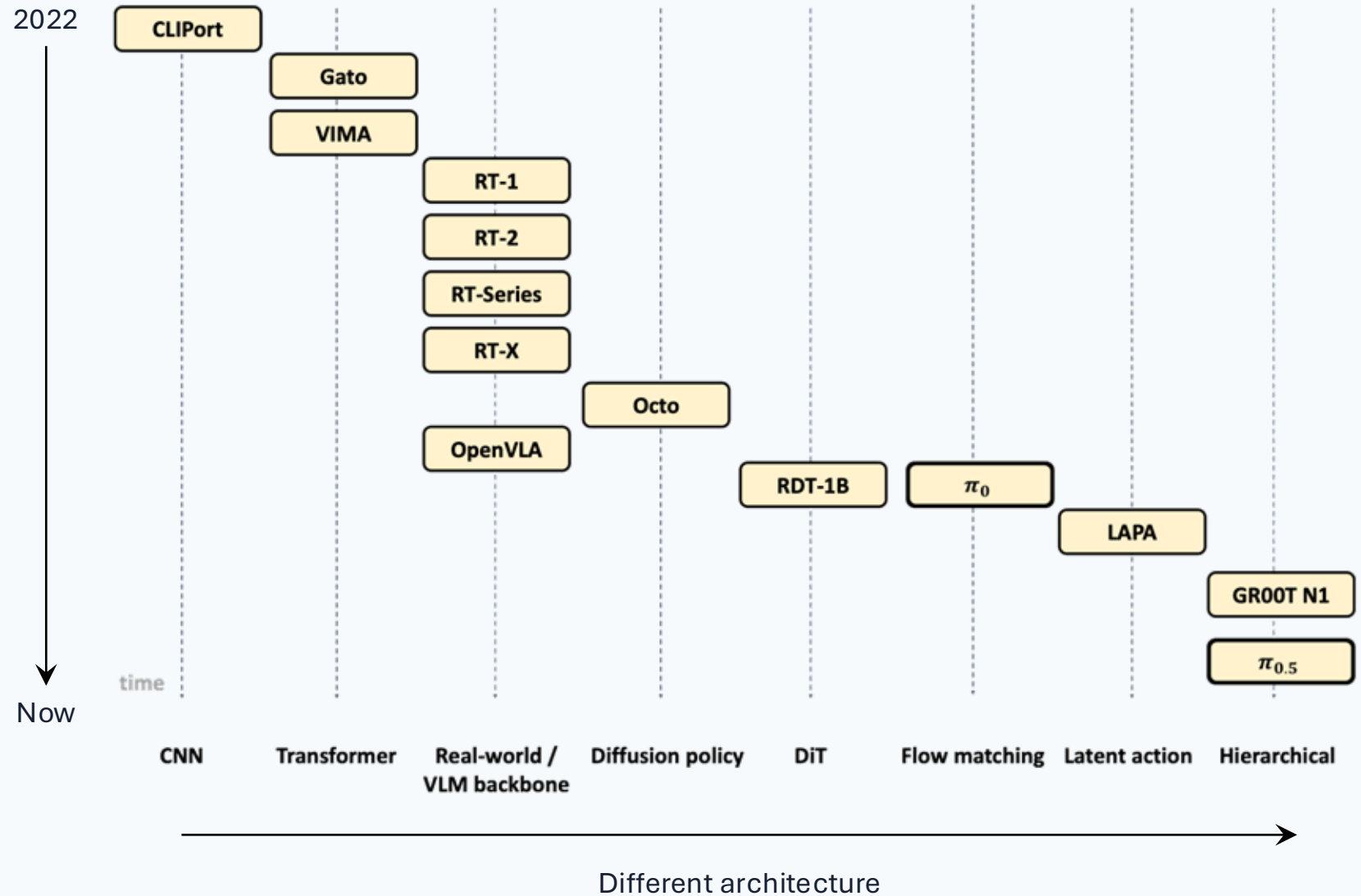
Challenges

- **Data availability:** datasets satisfying all three modalities are limited in both **scale** and **diversity**.
- **Embodiment mismatches:** each robot typically operates in a distinct **action space** and proprioceptive **observation space**, reflecting differences in degrees of freedom, sensor modalities, and kinematic structure.
- **Computational constraints:** training VLA models entails a considerable amount of computational demands due to the **high-dimensional** and **multi-modal** nature of their input.



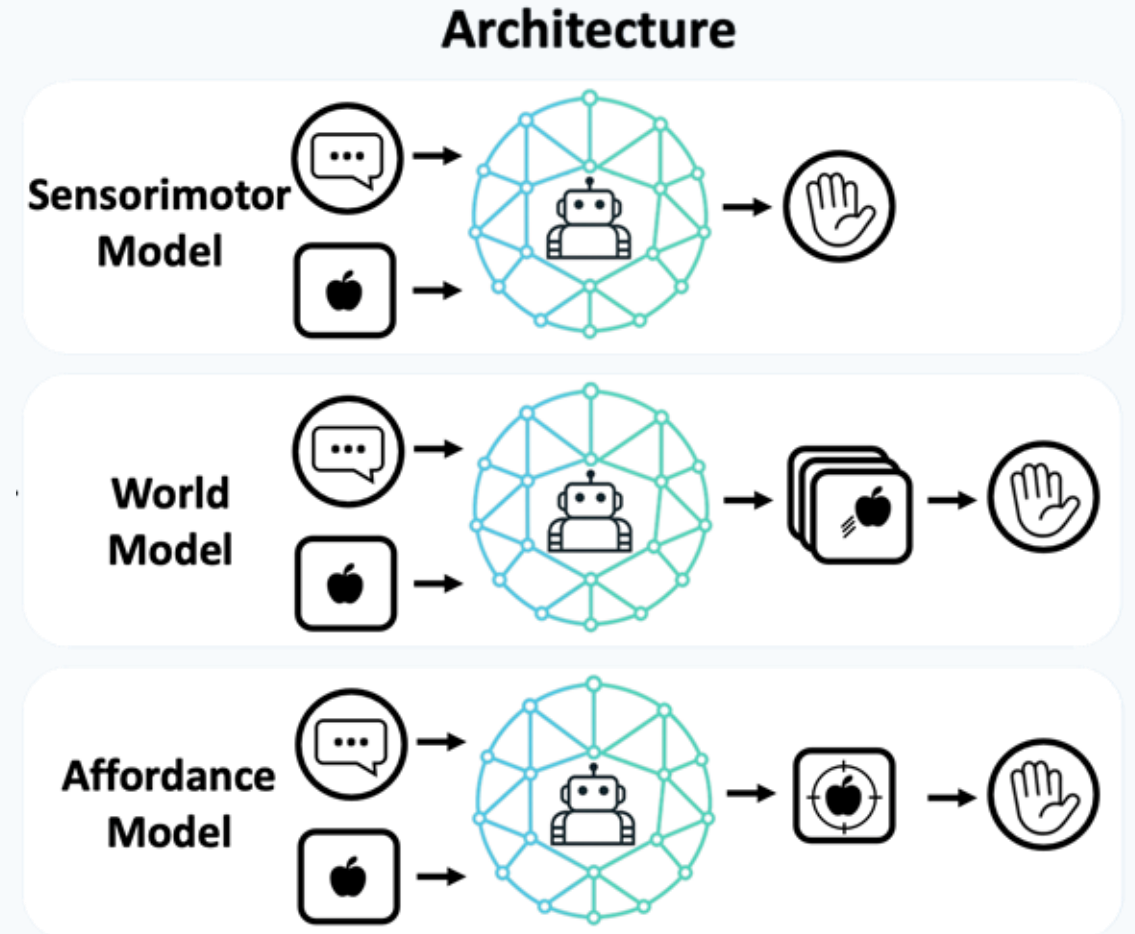
Current Status

- Research on VLAs remains in its early stages. Architectural and training methodologies are not yet standardized.



Architectures

- **Sensorimotor model:** directly output **actions**.
- **World model:** predict the future evolution of **visual** modalities and use these predictions to guide action generation
- **Affordance model:** predict action-relevant visual affordances based on **language** and then generate actions accordingly.

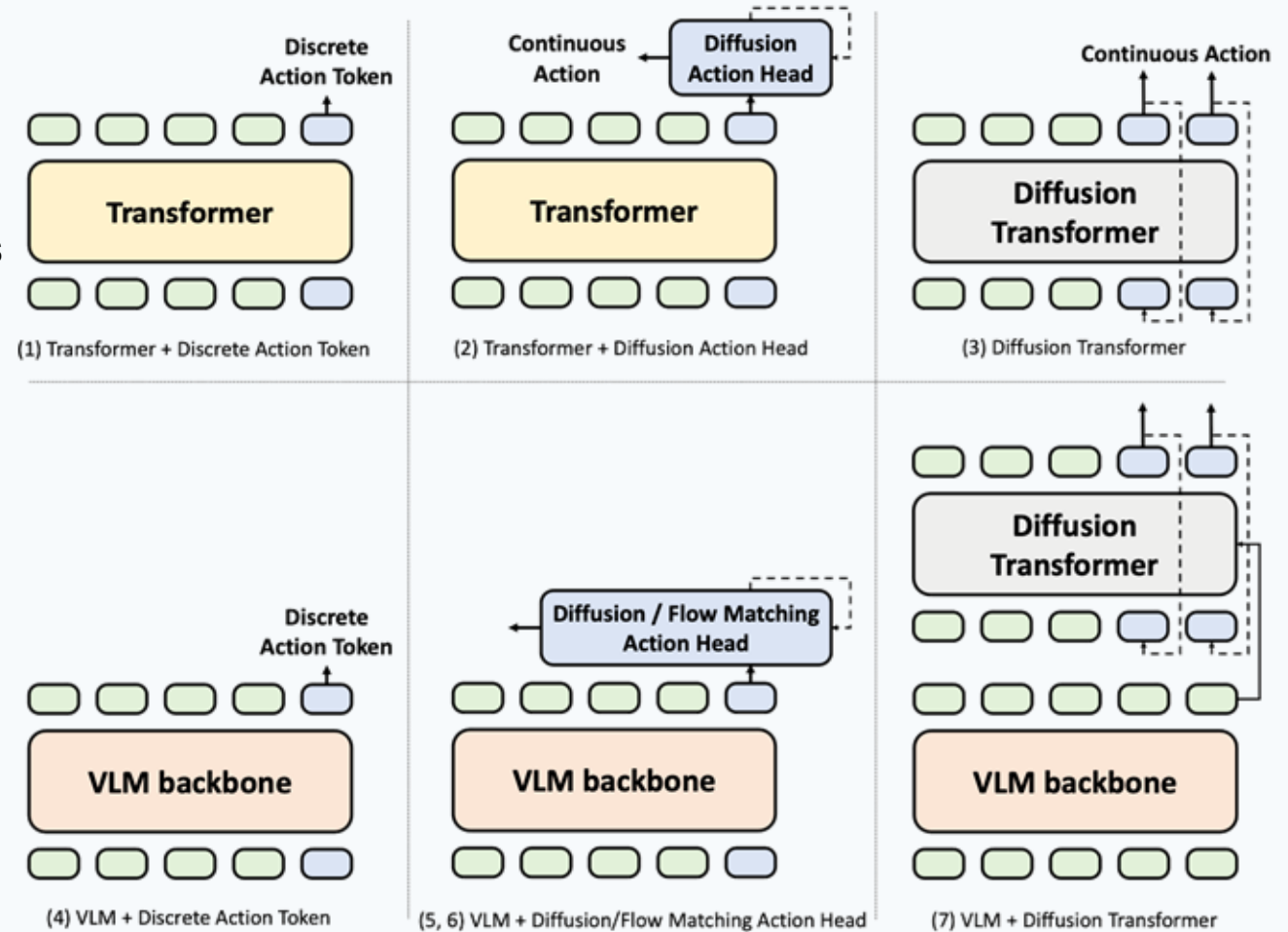


Architectures: Sensorimotor model

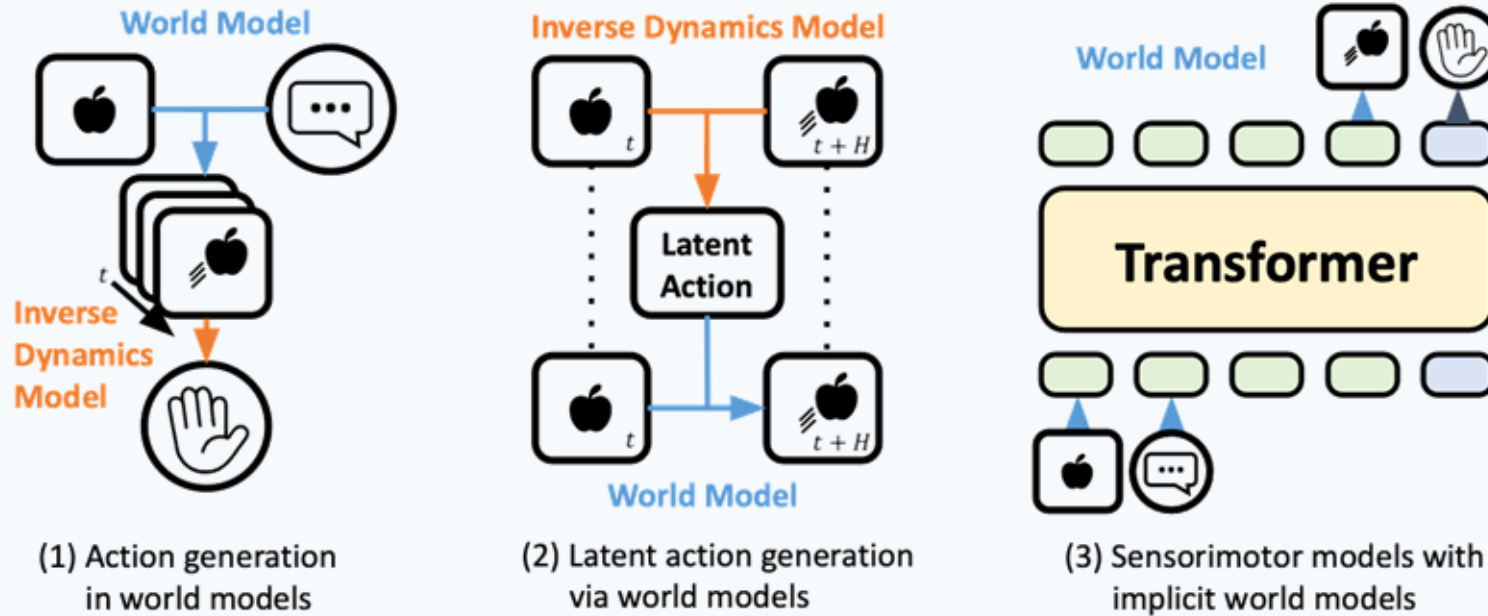
- From **Discrete Action Token** to **Continuous Action**: discrete action tokens often lack real-time responsiveness and smoothness, and diffusion models achieve continuous and stable action outputs.

- Leveraging **VLM** pre-trained on large-scale internet data: allows these models to incorporate human commonsense knowledge and benefit from in-context learning capabilities.

- VLM + Diffusion Transformer**: the VLM typically serves as a high-level policy, while the diffusion transformer acts as a low-level policy



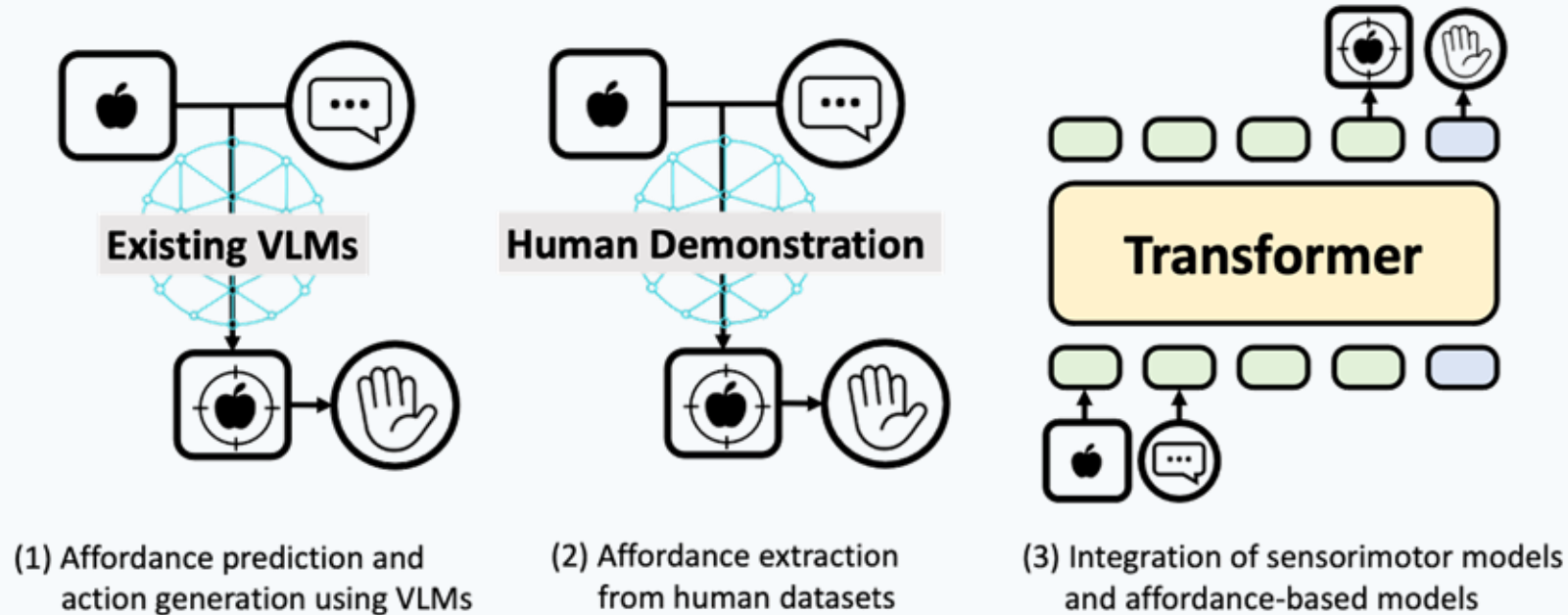
Architectures: World model



- **IDM** translates the predicted image sequence into low-level actions.
- Generate **future frames** conditioned on the current image and the **latent action**. (self-supervised)
- Jointly output both **actions** and predictions of **future observations** to improve performance

Architectures: Affordance model

*“An **affordance** is a relationship between an actor, an object, and an action, defining what actions are feasible for the actor on that object in a given environment.”*



- Predicting **affordances** and subsequently generating actions.
- Extracting affordances from **human demonstration videos** and learning latent representations to guide action generation
- Incorporating affordance prediction into VLA

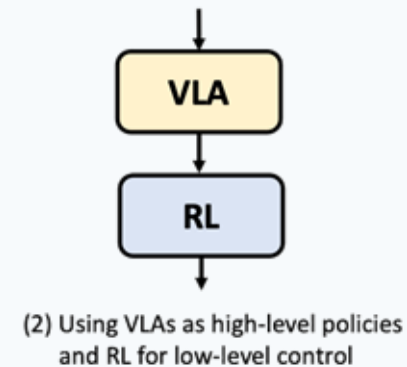
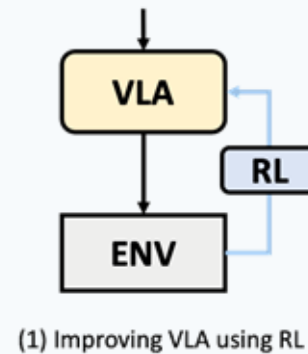
Architectures: Emerging directions

- **Hierarchical architectures:** Using existing VLMs as high-level policies to decompose task instructions into **subtasks**. These subtask descriptions are then passed to a VLA acting as the low-level policy.
- **Chain-of-Thought (CoT) reasoning:** introducing a step-by-step process between observations, instructions, and action generation, e.g. predicting **task descriptions**, **subtasks**, and **object positions**, before generating the final action sequence. (Broader concept)

Training strategy



- **Supervised learning:** training is often formulated as a next-token prediction task with pairs of images, language, and actions. Typically pre-training the VLM backbone and finetuning the action head on high-quality, robot- and task- specific data.
- **Self-supervised learning:** serving three main purposes, Modality alignment, Visual representation learning, Latent action representation learning.
- **Reinforcement learning:**



Data collection

Teleoperation



- Demonstrations are recorded in real time while a human operator directly controls the robot, enabling the collection of high-quality trajectories.

Proxy Devices



- More intuitive, flexible, and scalable data collection through the use of proxy devices. For example, a handheld gripper equipped with a GoPro camera.

Human Data Collection



- Collecting data by recording natural human behavior without relying on proxy devices.

Datasets

- **Human datasets:** more scalable, the advent of smart-glass-based recording devices has enabled more naturalistic and unobtrusive egocentric data collection.
- **Simulation datasets:** scalable, safe, and cost-effective.
- **Real robot datasets:**

Name	Episodes	Skill	Task	Modality	Embodiment	Collection
QT-Opt [336]	580K	1 (Pick)	NA	RGB	KUKA LBR iiwa	Learned
MT-Opt [337]	800K	2	12	RGB, L	7 robots	Scripted, Learned
RoboNet [338]	162K	NA	NA	RGB	7 robots	Scripted
BridgeData [339]	7.2K	4	71	RGB, L	WidowX 250	Teleop
BridgeData V2 [340]	60.1K	13	NA	RGB-D, L	WidowX 250	Teleop
BC-Z [341]	26.0K	3	100	RGB, L	Google EDR	Teleop
Language Table [329]	413K	1 (Push)	NA	RGB, L	xArm	Teleop
RH20T [335]	110K	42	147	RGB-D, L, F, A	4 robots	Teleop
RT-1 [16]	130K	12	700+	RGB, L	Google EDR	Teleop
OXE [17]	1.4M	527	160,266	RGB-D, L	22 robots	Mixed
DROID [330]	76K	86	NA	RGB-D, L	Franka	Teleop
FuSe [198]	27K	2	3	RGB, L, T, A,	WidowX 250	Teleop
RoboMIND [333]	107K	38	479	RGB-D, L	4 robots	Teleop
AgiBot World [94]	1M	87	217	RGB-D, L	AgiBot G1	Teleop

Diverse skills, tasks, robots; limited action modalities; mostly based on teleoperation

Application & Evaluation

Different robotic platforms



Benchmarks

- Evaluation metrics for VLA models remain poorly defined, particularly in real-world settings.
- Assessing generalization on physical robots is challenging due to differences in embodiment, safety concerns, and limited reproducibility.
- Most evaluations are conducted in simulation.

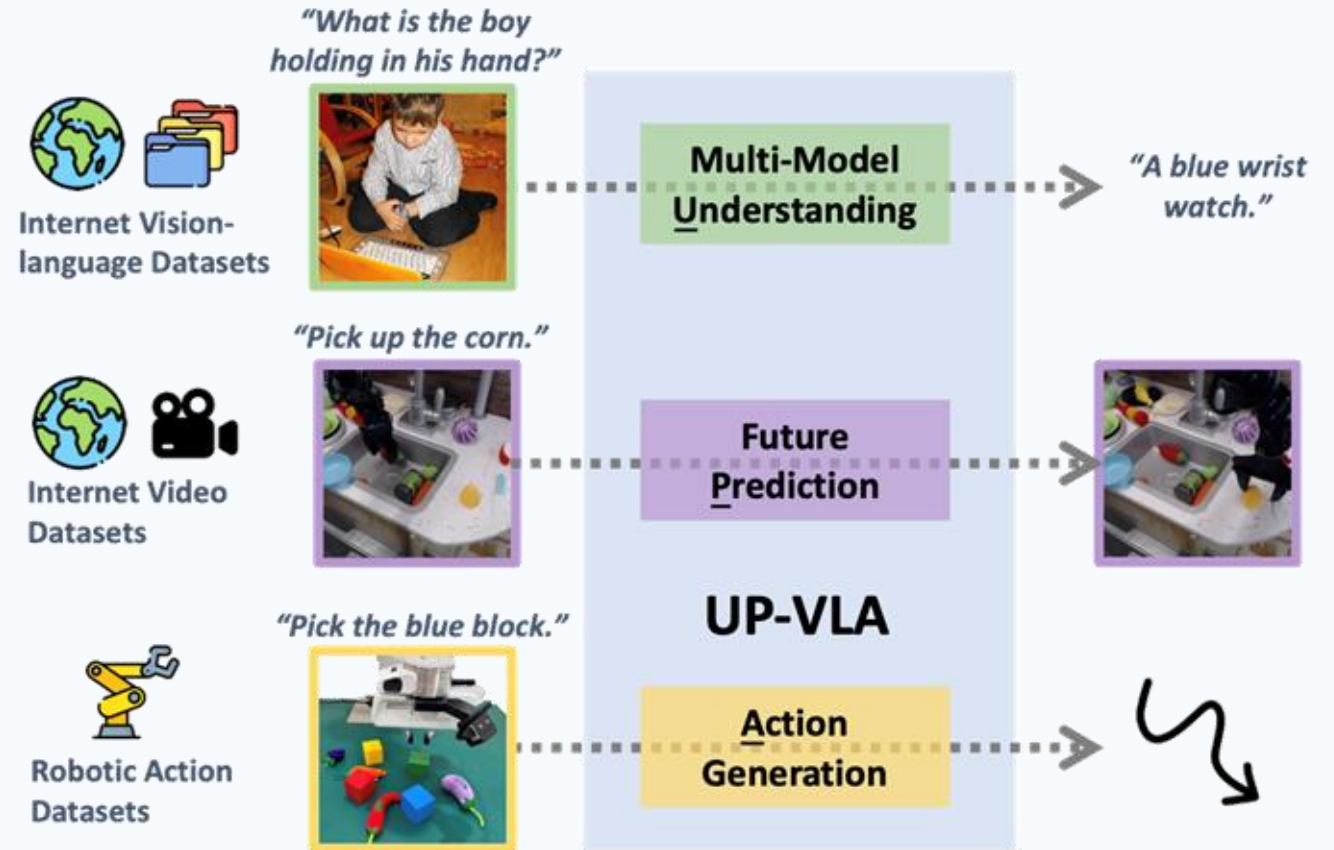
Name	Task	Scenes/Objects	Observation	Physics	Built Upon	Description
robosuite [381]	Manip	NA / 10	RGB-D, S	MuJoCo	NA	Modular framework, 11 tasks
robomimic [382]	Manip	NA / NA	RGB	MuJoCo	robosuite	Offline learning, 8 tasks
RoboCasa [383]	Manip	120 / 2.5K	RGB	MuJoCo	robosuite	100 kitchen tasks, photorealistic
LIBERO [343]	Manip	NA / NA	RGB	MuJoCo	robosuite	130 tasks in 4 task suites
Meta-World [384]	Manip	1 / 80	Pose	MuJoCo	NA	50 Manip tasks for Meta-RL
LeVERB-Bench [385]	Nav, WBC	4 / NA	RGB	PhysX	Isaac Sim	Humanoid control
ManiSkill [386]	Manip	NA / 162	RGB-D, PC, S	PhysX	SAPIEN	4 tasks, 36K demos
ManiSkill 2 [387]	Manip	NA / 2.1K	RGB-D, PC	PhysX	ManiSkill	Extended task diversity
ManiSkill 3 [388]	Nav, Manip, WBC	NA / NA	RGB-D, PC, S	PhysX	ManiSkill 2	GPU-parallelized simulation
ManiSkill-HAB [389]	Manip	105 / 92	RGB-D	PhysX	ManiSkill 3, Habitat 2.0	HAB tasks from Habitat 2.0
RoboTwin [390], [391]	Manip	NA / 731	RGB-D	PhysX	SAPIEN	Dual-arm tasks
Ravens [26]	Manip	NA / NA	RGB-D	PyBullet	NA	10 tabletop tasks
VIMA-BENCH [31]	Manip	NA / 29	RGB, S	PyBullet	Ravens	17 multimodal prompt tasks
LoHoRavens [392]	Manip	1 / 3	RGB-D	PyBullet	Ravens	Long-horizon planning
CALVIN [342]	Manip	4 / 7	RGB-D	PyBullet	NA	Long-horizon lang-cond tasks
Habitat [393]	Nav	185 / NA	RGB-D, S	Bullet	NA	Fast, Nav only
Habitat 2.0 [394]	Nav, Manip	105 / 92	RGB-D	Bullet	Habitat	Mobile manipulation (HAB)
Habitat 3.0 [395]	Nav, Manip	211 / 18K	RGB-D	Bullet	Habitat 2.0	Human avatars support
RLBench [396]	Manip	1 / 28	RGB-D, S	PyBullet	V-REP	Tiered task difficulty
THE COLOSSEUM [397]	Manip	1 / 107	RGB-D	PyBullet	RLBench	20 tasks, 14 env variations
AI2-THOR [398]	Nav, Manip	NA / 118	RGB-D, S	Unity	NA	Object states, task planning
CHORES [399]	Nav	191K / 40K	RGB	Unity	AI2-THOR	Shortest-path planning
SIMPLER [400]	Manip	4 / 17	RGB	PhysX	SAPIEN, Isaac Sim	Real-to-sim evaluation
RoboArena [401]	Manip	NA / NA	RGB	Real	NA	Distributed real-world evaluation

Future directions

- **More data modalities:** audio, tactile sensing, 3D point clouds, etc.
- **Long-horizon tasks:** reasoning, maintaining a form of memory, selectively attending to key information that supports decision-making.
- **Continual learning:** VLA systems remain fixed, making them vulnerable when faced with novel or out-of-distribution scenarios.
- **Safety:** current systems often lack mechanisms to detect and avoid unexpected human presence in the workspace, increasing the risk of collisions.
- **Failure detection:** current VLA systems lack mechanisms for detecting unexpected failures or responding appropriately.

UP-VLA: A Unified Understanding and Prediction Model for Embodied Agent

- **Motivation:** recent VLA models leverage VLMs, which focus on **high-level semantic** content and neglect **low-level features**, limiting their ability to capture detailed visual and spatial information.
- **What to unify and why:** a **Unified** VLA model training with both *multi-modal Understanding* and *future Prediction* objectives, enhancing both **high-level semantic** comprehension and **low-level spatial understanding**.



VLM-based model, Prediction-based model vs. UP-VLA

• VLM-based model:

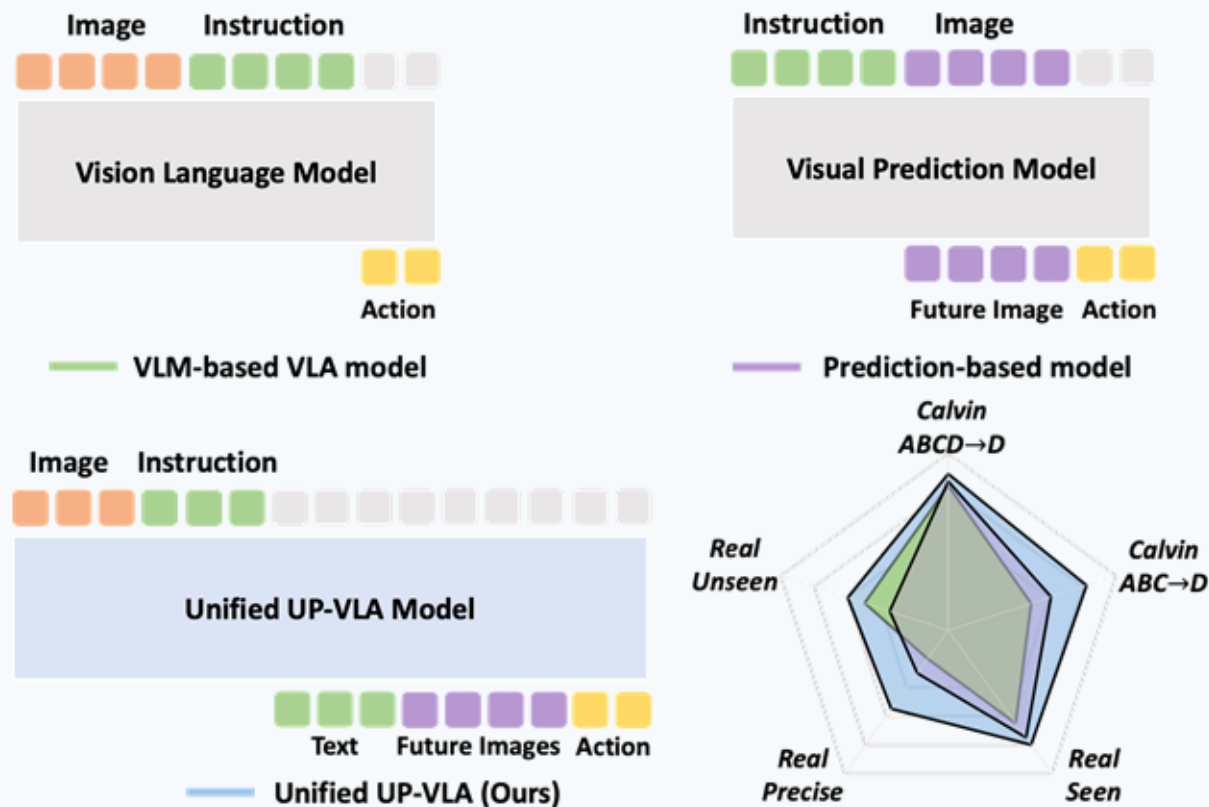
- ✓ in-distribution multitask settings
- ✓ real-world generalization task
- ✗ adaptation and precise control

• Prediction-based model (World model):

- ↑ adaptation and precise control

• UP-VLA:

- ✓ in-distribution multitask settings
- ✓ real-world generalization task
- ✓ adaptation and precise control

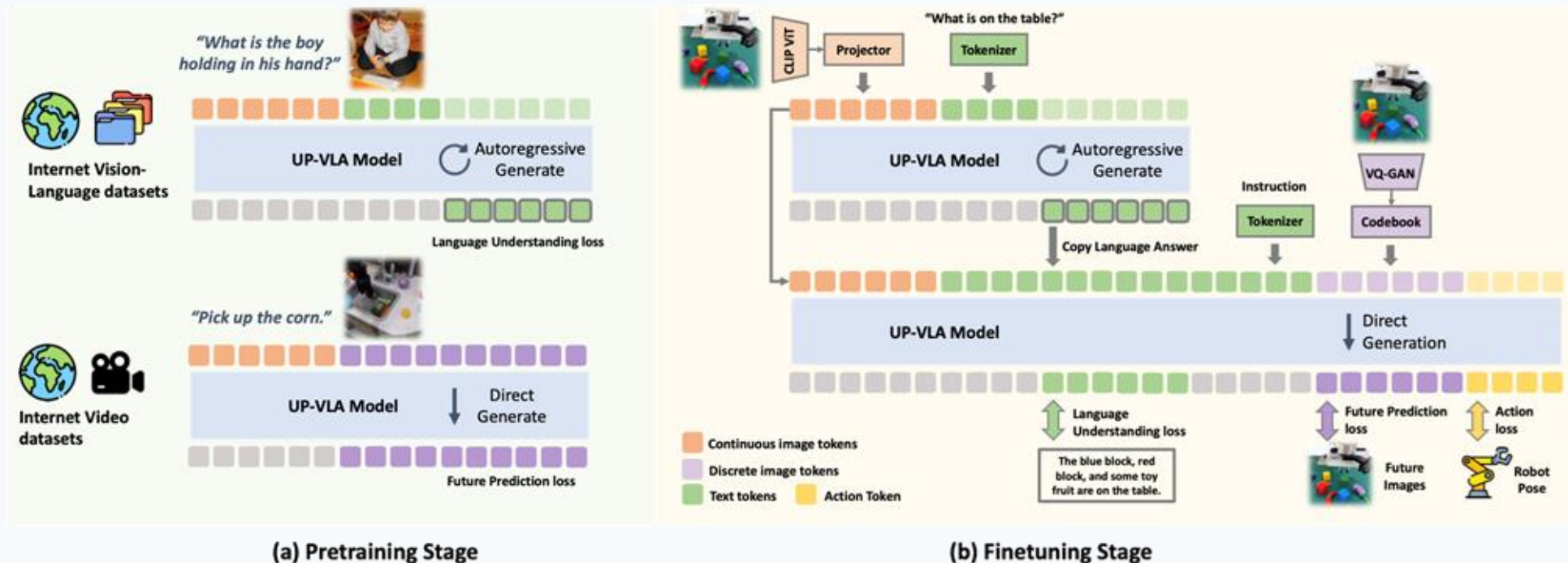


ABCD→D, Real Seen: in-distribution multitask learning capabilities

Real unseen: real-world semantic generalization

ABC→D, Real Precise: adaptation and precise control abilities

UP-VLA framework



- **Multi-modal Understanding:**

Given a paired image-text question-answering set (I, L) :

$$\hat{L} = \pi_{\theta}^{MMU}(I, L)$$

- **Future Visual Prediction:**

Given an image and instruction pair (O_t, L) at time t :

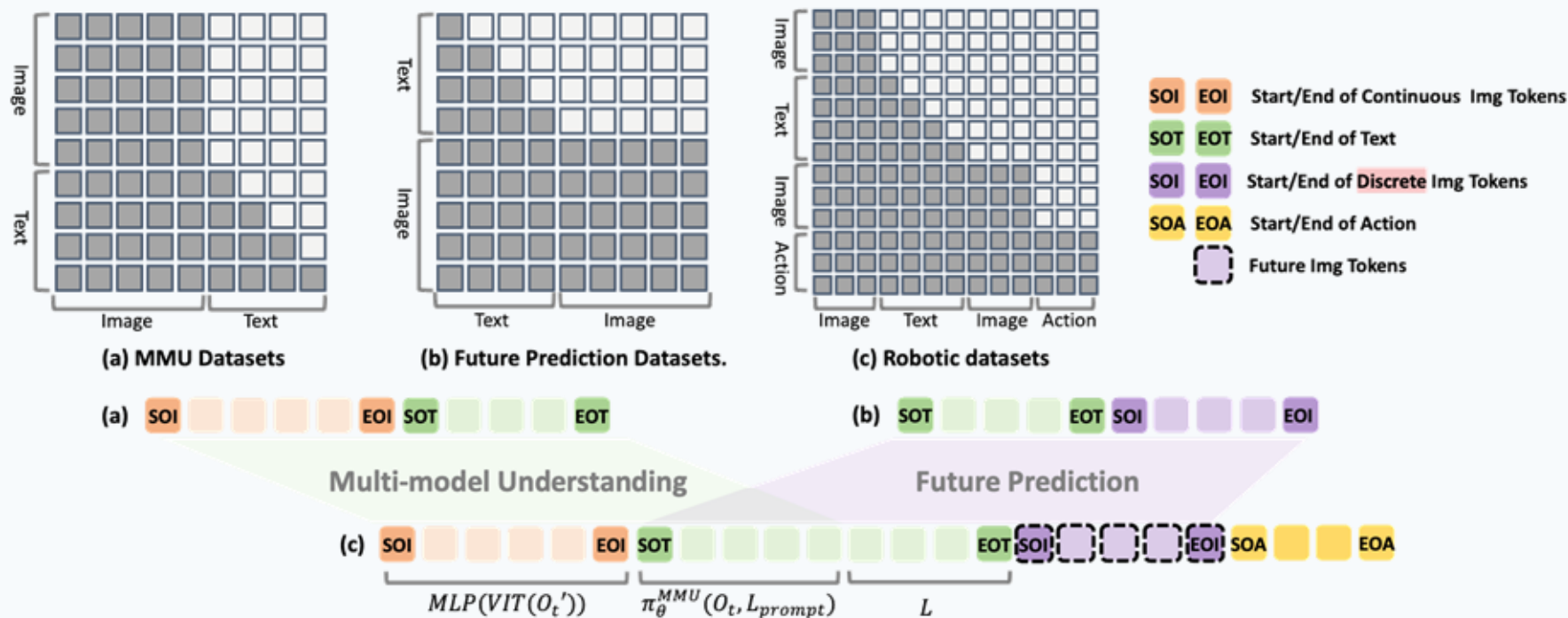
$$\hat{O}_{t+\Delta t} = \pi_{\theta}^{PRE}(O_t, L)$$

- **Enhancing Action Learning with Joint Prediction and Understanding:**

$$L' = [E_1(O'_t), \pi_{\theta}^{MMU}(O_t, L_{\text{prompt}}), L]$$

$$(\hat{O}_{t+\Delta t}, \hat{A}_{t:t+\Delta t}) = \pi_{\theta}^{PRE}(O_t, L')$$

Implementation details



Multi-modal Understanding:

- Continuous image token for better understanding
- Image token before text token: focusing on the image while comprehending the language

Future Visual Prediction:

- Discrete image token for better image reconstruction.
- Image token after text token: enabling the image to attend to all input information

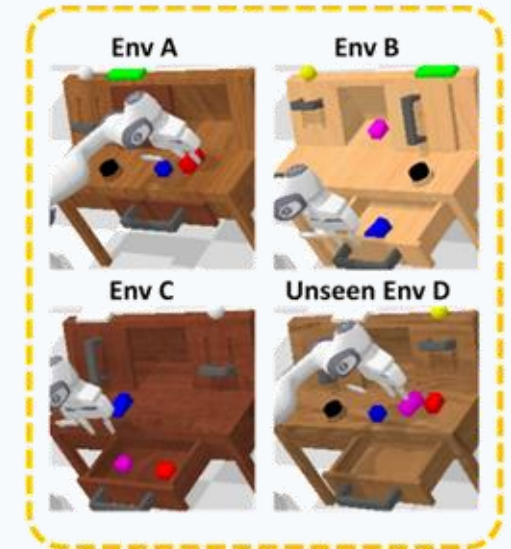
Jointly learning:

- Tokens from both tasks are concatenated, allowing actions to attend to both high-level scene understanding and low-level visual information

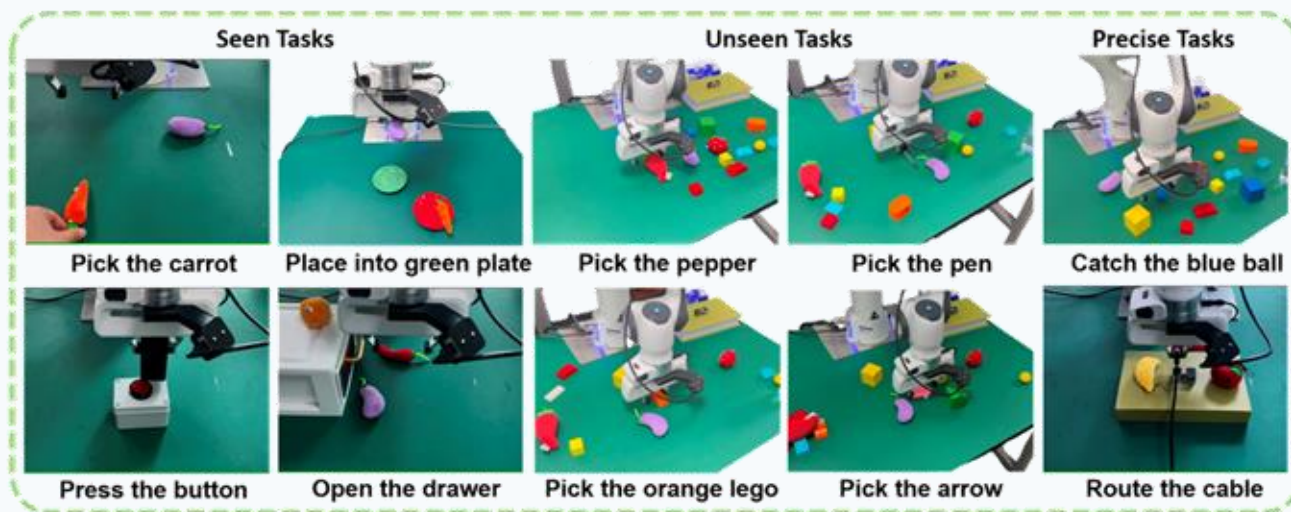
Simulation evaluation on Calvin benchmark

Method	Type	Task	Tasks completed in a row					Avg. Len $\uparrow$
			1	2	3	4	5	
RT-1	other	ABC $\rightarrow$ D	0.533	0.222	0.094	0.038	0.013	0.90
Diffusion Policy*	other	ABC $\rightarrow$ D	0.402	0.123	0.026	0.008	0.000	0.56
3D Diffuser Actor	other	ABC $\rightarrow$ D	0.938	<u>0.803</u>	<u>0.662</u>	<u>0.533</u>	0.412	<u>3.35</u>
3D-VLA	VLA	ABC $\rightarrow$ D	0.447	0.163	0.081	0.016	0.000	0.71
UP-VLA-RT-2*	VLA	ABC $\rightarrow$ D	0.612	0.389	0.236	0.138	0.062	1.44
Robo-Flamingo	VLA	ABC $\rightarrow$ D	0.824	0.619	0.466	0.331	0.235	2.47
Uni-Pi	Prediction	ABC $\rightarrow$ D	0.560	0.160	0.080	0.080	0.040	0.92
Susie	Prediction	ABC $\rightarrow$ D	0.870	0.690	0.490	0.380	0.260	2.69
GR-1	Prediction	ABC $\rightarrow$ D	0.854	0.712	0.596	0.497	0.401	3.06
UP-VLA-phi-w/o-mmu*	Prediction	ABC $\rightarrow$ D	0.844	0.705	0.604	0.520	<u>0.430</u>	3.13
UP-VLA	Prediction&VLA	ABC $\rightarrow$ D	<u>0.928</u>	0.865	0.815	0.769	0.699	4.08

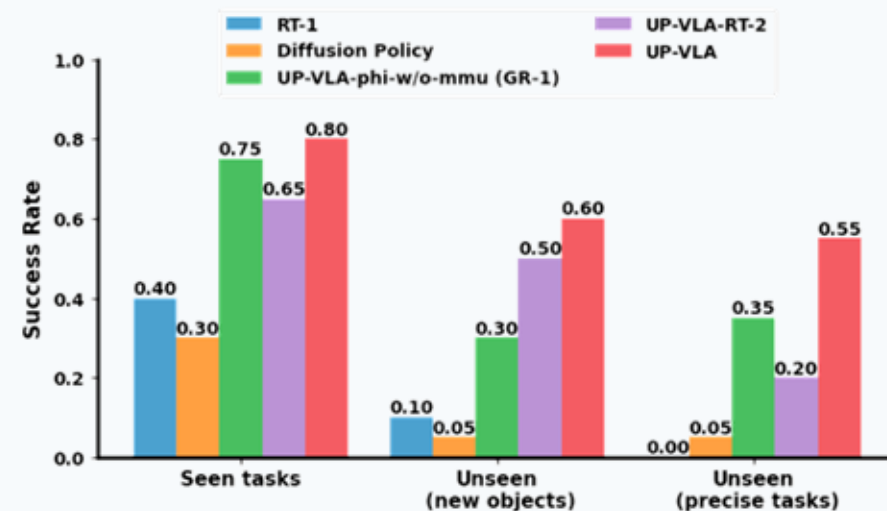
Method	Type	Task	Tasks completed in a row					Avg. Len $\uparrow$
			1	2	3	4	5	
RT-1	other	ABCD $\rightarrow$ D	0.844	0.617	0.438	0.323	0.227	2.45
Robo-Flamingo	VLA	ABCD $\rightarrow$ D	0.964	<u>0.896</u>	0.824	0.740	0.660	4.09
GR-1	Prediction	ABCD $\rightarrow$ D	0.949	<u>0.896</u>	<u>0.844</u>	<u>0.789</u>	<u>0.731</u>	<u>4.21</u>
UP-VLA	Prediction&VLA	ABCD $\rightarrow$ D	<u>0.962</u>	0.921	0.879	0.842	0.812	4.42



Real robot evaluation on Franka-Emika Panda robot



over 2k demonstrations above 6 skills



Takeaways

- A unified training strategy
- High-level semantic content and low-level visual and spatial details are both crucial for accurate robot control.
- Jointly modeling understanding and future image prediction helps.